Deep Dive: Literature Review

of Reference Papers

Project Context: Architecture-Agnostic Layer-Wise PTQ of LLMs (with a focus on Hessian-Diagonal Mixed-Precision Allocation).

This document summarizes the 9 research papers located in the papers/ directory, detailing their core contributions and how they directly relate to our project's methodology.

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01 GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers

FOUNDATION

File: 1 • Frantar et al., ICLR 2023

WHAT IT IS

The foundational paper that introduced GPTQ. It leverages approximate second-order information (the inverse Hessian matrix) to perform one-shot weight quantization, processing weights column-by-column using the Optimal Brain Surgeon (OBS) update rule.

CONNECTION TO OUR PROJECT

This is the baseline algorithm we are expanding upon. While GPTQ assumes a uniform bit-width across the model, our Phase 2 introduces “Hessian-Diagonal Mixed-Precision” to dynamically allocate bits based on the very Hessian matrices GPTQ computes.

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02 Benchmarking Post-Training Quantization in LLMs

PHASE 1

File: 2.pdf • Zhao et al., 2025

WHAT IT IS

A massive benchmarking study that categorizes PTQ into four strategies: Compensation (e.g., GPTQ), Rotation (e.g., QuIP), Salience (e.g., AWQ), and Optimization. It evaluates their cross-structure (Mamba, MoE) and cross-bitwidth robustness.

CONNECTION TO OUR PROJECT

Directly aligns with our Phase 1. We are also building an ablation grid across different architectures (Qwen, LLaMA, Pythia) and methods (RTN, GPTQ, Hadamard). Their findings validate our approach of evaluating cross-architecture PTQ resilience.

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03 APTQ: Attention-aware Post-Training Mixed-Precision Quantization

PHASE 2

File: 3.pdf • Guan et al., DAC 2024

WHAT IT IS

Introduces APTQ, which expands second-order quantization to consider the entire attention block (including non-linear softmax). Crucially, it proposes a Hessian trace-driven mixed-precision quantization scheme.

CONNECTION TO OUR PROJECT

Highly relevant to Phase 2. They use the trace of the Hessian to assign 2-bit or 4-bit precision to layers. We are doing something very similar, but using the diagonal of the Hessian (H_jj) multiplied by weight variance as our sensitivity metric.

04

GAMMA: Global Bit Allocation for Mixed-Precision Models

PHASE 2

File: 4.pdf • Yao et al., 2026

WHAT IT IS

Proposes a quantizer-agnostic framework for mixed-precision allocation that avoids the massive compute costs of search-based methods or the inaccuracy of static proxy metrics. It learns module-wise bit preferences via a teacher-forced hidden-state reconstruction objective.

CONNECTION TO OUR PROJECT

Represents the state-of-the-art alternative to Hessian-based mixed precision. While we use a closed-form analytical score (Hessian-diagonal), GAMMA uses a differentiable learning pipeline. It is an excellent comparison point for our Phase 2 literature review.

05

KRONQ: LLM Quantization via Kronecker-Factored Hessian

EXTENSION

File: 5.pdf • Lee et al., 2026

WHAT IT IS

Challenges the standard GPTQ assumption that only input activation statistics matter. KRONQ uses Kronecker-factored approximations to include gradient covariance (output-side statistics) in the quantization objective and mixed-precision allocation.

CONNECTION TO OUR PROJECT

Extremely relevant mathematical extension of GPTQ. While we use the activation Hessian (H = X^T X), they prove that incorporating the gradient Hessian (H_G) improves 2-bit performance on massive models.

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06 D²Quant: Accurate Low-bit Post-Training Weight Quantization

SUPPORTING

File: 6.pdf • Yan et al., 2026

WHAT IT IS

Identifies that sub-4-bit PTQ degrades due to down-projection matrices and activation mean-shifts. It introduces a Dual-Scale Quantizer (DSQ) for weights and a Deviation-Aware Correction (DAC) that adjusts biases in LayerNorm to fix activation drift.

CONNECTION TO OUR PROJECT

Provides a structural explanation for why GPTQ fails at 2-bit. In our project, if we notice severe degradation at 2-bit (e.g., in Qwen2.5), we can cite D²Quant to explain that LayerNorm activation drift is the likely culprit.

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07 Task-Stratified Knowledge Scaling Laws for PTQ

SUPPORTING

File: 7.pdf • Zhou et al., 2026

WHAT IT IS

Investigates how fine-grained PTQ factors (group size and calibration set size) impact different types of LLM knowledge (memorization, application, reasoning). Finds that reasoning is highly sensitive to bit-width, while memorization is sensitive to calibration data.

CONNECTION TO OUR PROJECT

Validates our use of WikiText-2 as a calibration set. If we see that our 2-bit models fail on reasoning tasks (like WinoGrande) but survive on easier tasks, this paper provides the theoretical backing for that phenomenon.

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08 CrossQuant: A PTQ Method with Smaller Quantization Kernel

BACKGROUND

File: 8.pdf • Liu et al., 2024

WHAT IT IS

Defines the “quantization kernel” as the set of elements in an activation matrix that get quantized to exactly zero. Proves that rounding small values to zero is the root cause of precision loss, and proposes cross-quantizing using both row and column maximums.

CONNECTION TO OUR PROJECT

Provides a deep mathematical look at activation quantization. While our project is primarily focused on weight-only quantization, this paper is useful background reading on why outlier features distort LLM mathematics.

09

Adapting Methods for Domain-Specific Japanese Small LMs

PHASE 1

File: 9.pdf • Yasuno, 2026

WHAT IT IS

A practical engineering paper about fine-tuning Japanese LLMs via QLoRA, and evaluating how 4-bit quantization impacts them. It makes a notable finding that GQA (Grouped-Query Attention) architectures degrade severely under 4-bit quantization compared to standard MHA.

CONNECTION TO OUR PROJECT

Directly supports our cross-architecture hypothesis in Phase 1! In our ablation grid, Qwen2.5 uses GQA while Pythia uses MHA. If our grid shows Qwen2.5 struggling at 2/3-bit compared to Pythia, we can use Yasuno's findings on GQA vulnerability to support our conclusions.